Project: Worked examples
Title: Tolerance stack-up (1D) v2
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

A linear tolerance loop: worst-case limits, root-sum-square and modified RSS limits, predicted spread and capability against a stated requirement, contribution Pareto and a Monte Carlo prediction.
Worst case: Σ|half-bands|. RSS: √Σ half². Modified RSS: factor × RSS (Bender-style). Predicted σ: √Σ σ_i² with σ_i from each contributor's distribution. C_pk from the nearer limit. Monte Carlo: seeded sampling of every contributor, summed with its sign.
Ref: BS 8888 — Technical product documentation and specification
Ref: ASME Y14.5 — Dimensioning and tolerancing
Ref: ISO 2768-1 — General tolerances (where a contributor's band is a general tolerance)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
contributors	Contributors in the loop	4 row(s)		spec: Shigley 9e Ch. 1 example, part dimensions	any
contributors[1]		name = a shoulder length; direction = adds; nominal = 1.75 in; upper = 0.003 in; lower = -0.003 in			
contributors[2]		name = b sleeve; direction = subtracts; nominal = 0.75 in; upper = 0.001 in; lower = -0.001 in			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
contributors[3]		name = c washer; direction = subtracts; nominal = 0.12 in; upper = 0.005 in; lower = -0.005 in			
contributors[4]		name = d sleeve; direction = subtracts; nominal = 0.875 in; upper = 0.001 in; lower = -0.001 in			
minimum	Minimum acceptable result	0.003	in	spec: Shigley 9e Ch. 1 example, required gap	-39370 … 39370 in
k	Standard deviations per half-band (normal contributors)	3		assumption: Each part at ±3σ within its band.	1 … 6
f_rss	Modified-RSS factor (Bender-style)	1.5		assumption: Bender factor 1.5.	1 … 2
cpk_target	Required C_pk	1.33		assumption: Usual capability target.	1 … 2
n_mc	Monte Carlo samples	10000		assumption: Default sample count.	1000 … 100000
seed	Monte Carlo seed	1		assumption: Fixed seed for reproducibility.	0 … 4295000000
notation	Drawing notation	ASME Y14.5		assumption: Inch drawing.	any

3. Assumptions

A1. Every contributor acts along the one axis of the stack. (A linear stack: no angular or datum-shift effects beyond what the user enters as a contributor.)
A2. Statistical figures assume contributors vary independently about their mid-limit (plus any stated mean shift), each with the distribution stated for it. (The worst case is a guarantee; every statistical figure is a prediction about a population.)
A3. A position or profile contributor contributes ± half its tolerance zone, plus half the stated MMC bonus. (Diametral zones are bilateral about the true position.)
A4. k (Standard deviations per half-band (normal contributors)) is taken as 3. (Each part at ±3σ within its

band.)
A5. f_rss (Modified-RSS factor (Bender-style)) is taken as 1.5. (Bender factor 1.5.)
A6. cpk_target (Required C_pk) is taken as 1.33. (Usual capability target.)
A7. n_mc (Monte Carlo samples) is taken as 10000. (Default sample count.)
A8. seed (Monte Carlo seed) is taken as 1. (Fixed seed for reproducibility.)
A9. notation (Drawing notation) is taken as ASME Y14.5. (Inch drawing.)

4. Working

```
[6] c1 = + 1.75 +0.003/-0.003
= mid 1.75 in, ± 0.003 in, □ 0.001 in (normal)
= 1.75 in
```
a shoulder length
```
[7] c2 = - 0.75 +0.001/-0.001
= mid 0.75 in, ± 0.001 in, □ 0.000333333 in (normal)
= -0.75 in
```
b sleeve
```
[8] c3 = - 0.12 +0.005/-0.005
= mid 0.12 in, ± 0.005 in, □ 0.00166667 in (normal)
= -0.12 in
```
c washer
```
[9] c4 = - 0.875 +0.001/-0.001
= mid 0.875 in, ± 0.001 in, □ 0.000333333 in (normal)
= -0.875 in
```
d sleeve
```
[10] x_nom = □ (sign × nominal)
= 1.75 - 0.75 - 0.12 - 0.875
= 0.005 in
```
Nominal result
```
[11] x_mean = □ (sign × mid-limit)
= 1.75 - 0.75 - 0.12 - 0.875
= 0.005 in
```
Mean result
```
[12] t_wc = □ half-bands
= 0.003 + 0.001 + 0.005 + 0.001
= 0.01 in
```
Worst-case tolerance (±)
```
[13] x_wc_min = x_mean - t_wc
= 0.005 in − 0.01 in
= -0.005 in
```
Worst-case minimum
```
[14] x_wc_max = x_mean + t_wc
= 0.005 in + 0.01 in
= 0.015 in
```
Worst-case maximum
```
[15] t_rss = √□ half²
= √(0.003² + 0.001² + 0.005² + 0.001²)
= 0.006 in
```
Root-sum-square tolerance (±)
```
[16] x_rss_min = x_mean - t_rss
= 0.005 in − 0.006 in
= -0.001 in
```
RSS minimum

[17] x_rss_max = x_mean + t_rss
= 0.005 in + 0.006 in
= 0.011 in
RSS maximum
[18] t_mrss = f_rss * t_rss
= 1.5 × 0.006 in
= 0.009 in
Modified RSS tolerance (±)
[19] sigma = √□ □_i²
= √(0.001² + 0.000333333² + 0.00166667² + 0.000333333²)
= 0.002 in
Predicted standard deviation of the result
[20] m_min = x_wc_min - minimum
= -0.005 - 0.003
= -0.008 in
Worst-case margin to the minimum
[21] cpk = min(x_mean - LSL, USL - x_mean) / (3 □)
= min(0.002, ∞) / (3 × 0.002)
= 0.33333
Process capability index C_pk
[22] dpmo = 10⁶ × P(outside)
= 10⁶ × □-tails at □ = 0.002
= 158660
Predicted defects per million
[23] yield = 1 - dpmo / 10⁶
= 1 - 158660 / 10⁶
= 84.134 %
Predicted yield
[24] u_cpk = cpk_target / cpk
= 1.33 / 0.33333
= 3.99
Capability utilisation
[25] pareto = largest contributor (statistical)
= a shoulder length 25 %; b sleeve 2.78 %; c washer 69.4 %; d sleeve 2.78 %
= "c washer"
Contribution Pareto
[28] mc_mean = mean of 10000 samples (seed 1)
= Monte Carlo
= 0.0050019 in
Monte Carlo mean
[29] mc_sigma = sample standard deviation
= Monte Carlo
= 0.0019919 in
Monte Carlo σ
[30] mc_p05 = 5th percentile
= Monte Carlo
= 0.0017324 in
Monte Carlo 5 %
[31] mc_p95 = 95th percentile
= Monte Carlo
= 0.0082769 in
Monte Carlo 95 %

```
[32] mc_yield = samples inside the requirement
= 8461 of 10000
= 84.61 %
```
Monte Carlo yield

5. Results

SYMBOL	VALUE	DESCRIPTION
x_nom	0.005 in	Nominal result
x_mean	0.005 in	Mean result
t_wc	0.01 in	Worst-case tolerance (±)
x_wc_min	-0.005 in	Worst-case minimum
x_wc_max	0.015 in	Worst-case maximum
t_rss	0.006 in	RSS tolerance (±)
t_mrss	0.009 in	Modified RSS tolerance (±)
sigma	0.002 in	Predicted σ
m_min	-0.008 in	Worst-case margin to the minimum
cpk	0.33333	C_pk
dpmo	158660	DPMO (predicted)
yield	84.134 %	Yield (predicted)
mc_mean	0.0050019 in	Monte Carlo mean
mc_sigma	0.0019919 in	Monte Carlo σ
mc_p05	0.0017324 in	Monte Carlo 5th percentile
mc_p95	0.0082769 in	Monte Carlo 95th percentile
mc_yield	84.61 %	Monte Carlo yield

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Worst case within requirement	-0.005 … 0.015 in	0.003 … open in	—	
FAIL	C_pk against target	u_cpk = 3.99	u_cpk ≤ 1	3.99	

7. Applicability

No warnings.

8. Verification

```
Template mech.tolerance-stack@2.1.0 Content hash 49bec2d079be76ee Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```